AI X CITY CLIMATE ACTION HACKATHON 2026

FloodFact AI
Flood warnings people can trust.
Rumours that stop spreading.

Concept — Phase 1 Pilot: Mukuru, Nairobi (kwa Reuben · kwa Njenga · Viwandani)

THE CITY CLIMATE PROBLEM

Every long rain, two disasters hit at once

50%

of Kibera households flooded in a single rainy season

World Economic Forum

72 hrs

average delay before official alerts reached residents

Kenya Red Cross, 2023

3 in 5

WhatsApp messages during the 2023 floods were unverified

iHub Kenya Research

The flood is the first emergency. The wave of unverified WhatsApp rumours that follows — false dam-burst warnings, panicked evacuations, then real alerts ignored because trust is gone — is the second. Nairobi's informal settlements hold 60% of the city's population but receive less than 10% of its climate adaptation resources.

THIS ALREADY HAPPENED

Real incidents, not hypotheticals

Mathare — April 2024

Flash floods struck at 2 AM with no official alert issued. A viral WhatsApp voice note falsely claiming a nearby dam had burst sent families fleeing into more dangerous streets — two people were injured in the panic.

Kenya Red Cross field report, Apr 2024

Kibera — May 2023

A KMD regional advisory was issued but never reached households — a 72-hour average delay. So many false WhatsApp warnings had already circulated that residents dismissed the eventual official SMS alerts too.

Kenya Red Cross 2023; iHub Kenya Research

THE GAP

Early-warning tools and fact-checkers each solve half the problem

Early-warning systems

Broadcast alerts from weather/hydrology data.

Gap: don't touch the disinformation that undermines trust in the alert itself.

Fact-checking tools

Verify viral claims after the fact.

Gap: not built for real-time, hyperlocal, life-safety decisions.

FloodFact AI

Cross-checks the report against real evidence AND classifies rumours — in the same pipeline, in seconds.

Built for exactly this gap.

THE IDEA

Community-centered early warning and verification, in one pipeline

Residents report. AI cross-checks against real evidence. Trusted alerts go back out through the channels people already use.

1

Community Submits

WhatsApp, SMS, the web, or a trained youth ambassador.

2

AI Cross-Checks

Rainfall, flood-risk zones, history, and ground truth — gathered in parallel.

3

Classified in Seconds

A deterministic, exhaustively tested engine decides — never an LLM guess.

4

Alert Delivered

Back out via WhatsApp, SMS, notice points, and youth ambassadors.

TARGET USER & COMMUNITY

Built around the people who live the problem

1 Residents

Submit a report or a suspicious message; get a clear verdict and action, in their language.

2 Youth Ambassadors

Trained community members who validate on the ground, explain results locally, and log ground-truth observations.

3 Ops / Admin Team

Monitor live activity on a mission-control dashboard, review escalations, watch data-source health.

4 Partners

KMD, Kenya Red Cross, UN-Habitat, youth groups, county government — coordinate response using the same evidence trail.

What happens inside "AI Cross-Checks"

1

Understa nd

Claude extracts hazard type, location, claims — never infers facts not stated.

2

Geocode

A hyperlocal gazetteer + PostGIS resolve the exact pilot sub-area.

3

Gather Evidence

Rainfall, flood-risk, history, corroboration, ambassador truth, rumour similarity — in parallel.

4

Decide

A pure, deterministic function scores the evidence. Not Claude.

5

Explain

Claude writes the plain-language, bilingual rationale for the decision already made.

6

Alert

Verdict-specific message dispatched to the right channels.

OUR UNIQUE INNOVATION

The decision is deterministic and tested — not an LLM's guess

Safety rule	What it prevents
Insufficient evidence → always escalate	A confident verdict from too little evidence, in either direction.
Missing data ≠ zero	Treating an unavailable reading as "no risk" instead of "unknown".
Corroboration is weighted, not decisive	A popular or repeated message manufacturing false confidence.
Conflicts are surfaced, not resolved silently	Picking a side when evidence genuinely disagrees.
"No evidence of flooding" ≠ "evidence of no flooding"	Calling a rumour false just because nothing was found yet.

23 unit tests in tests/unit/risk-engine.test.ts encode these rules directly — every one is checked on every commit via CI.

SECURITY & TRUST

Privacy and auditability are structural, not an afterthought

Contact isolation

Real phone numbers live in one table with no default read access — not even for admins. Revealing one is logged.

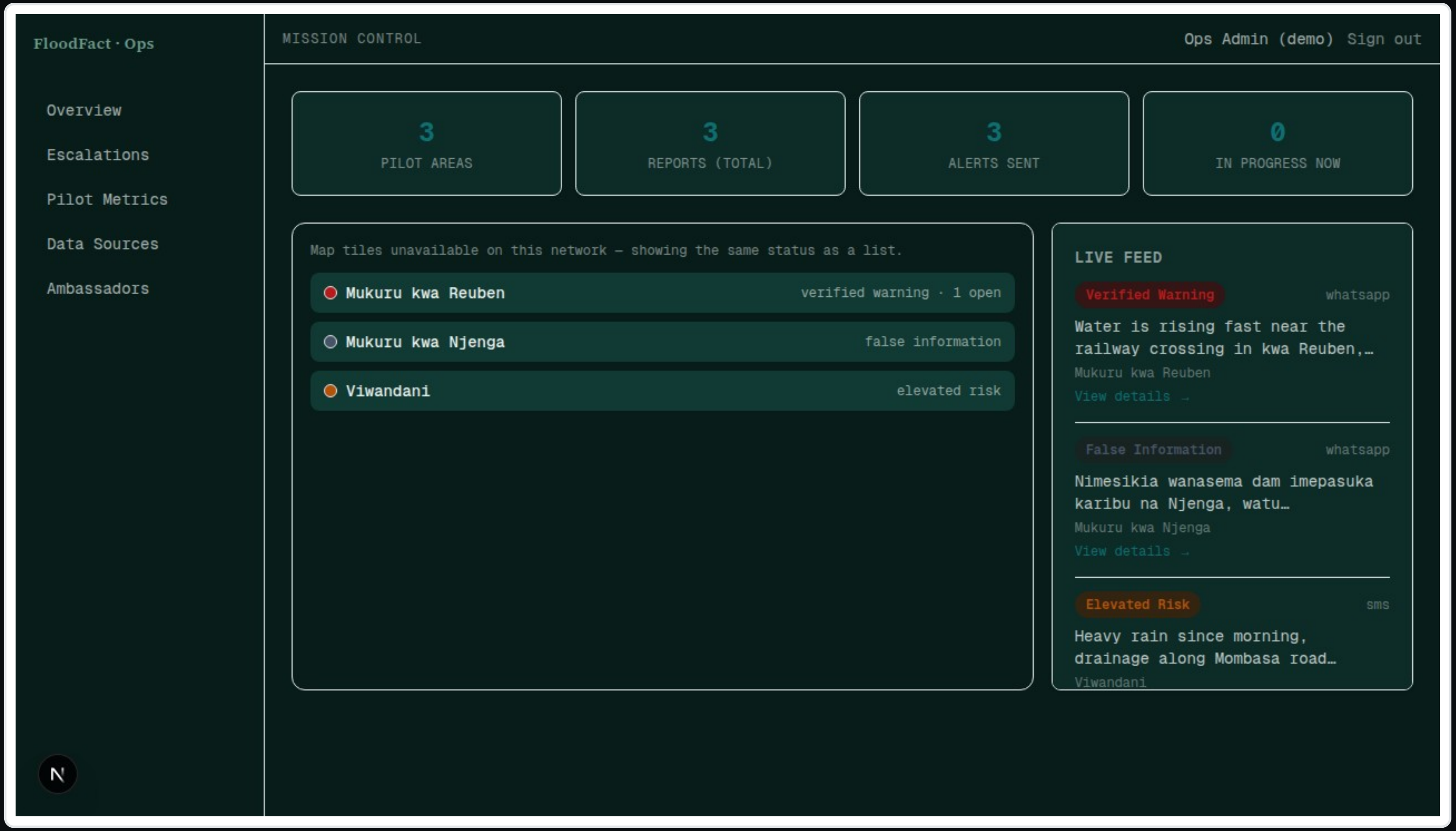
Row Level Security everywhere

Every table is RLS-scoped: ambassadors see only their ward; residents get no direct table access at all.

Full audit trail

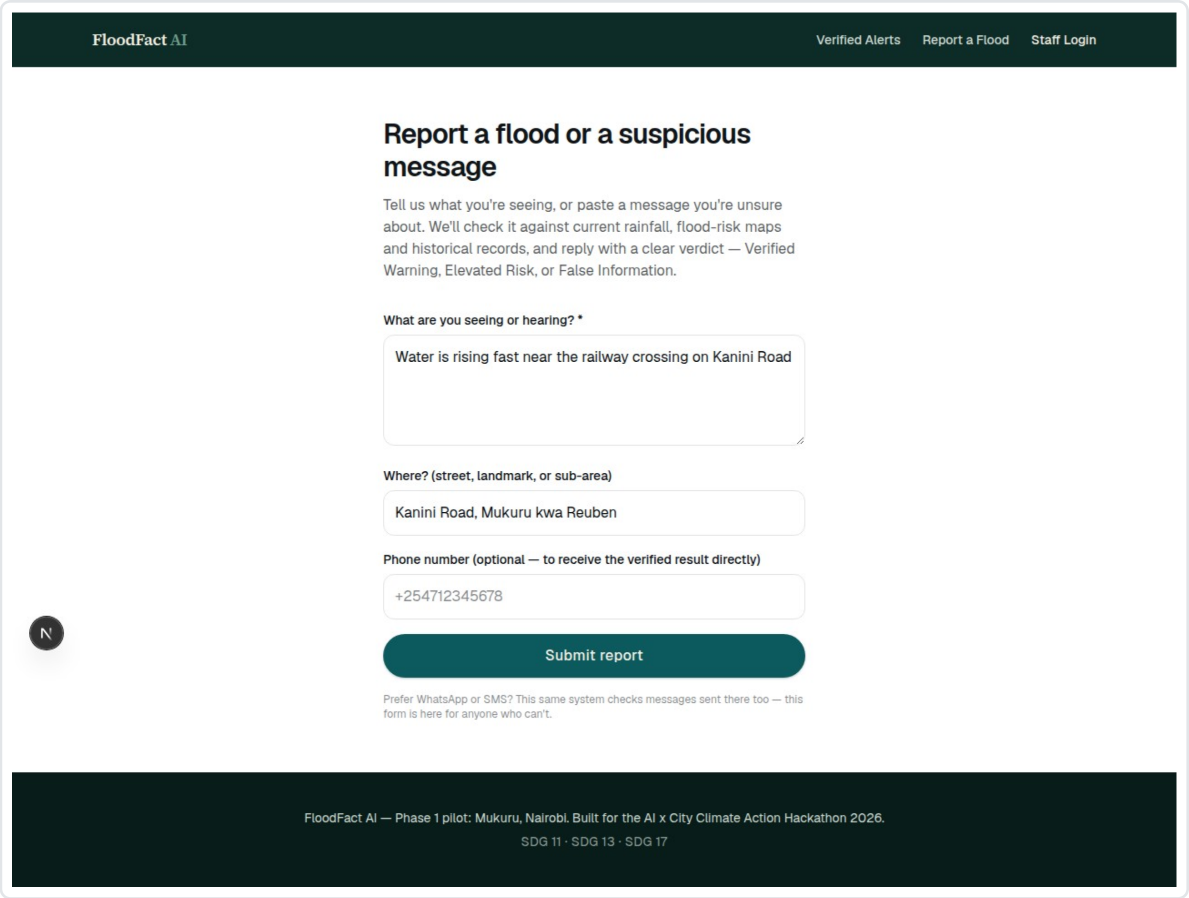
Every classification and alert is traceable to the evidence snapshot and model version that produced it.

Admins see every report, live, on a real-time map



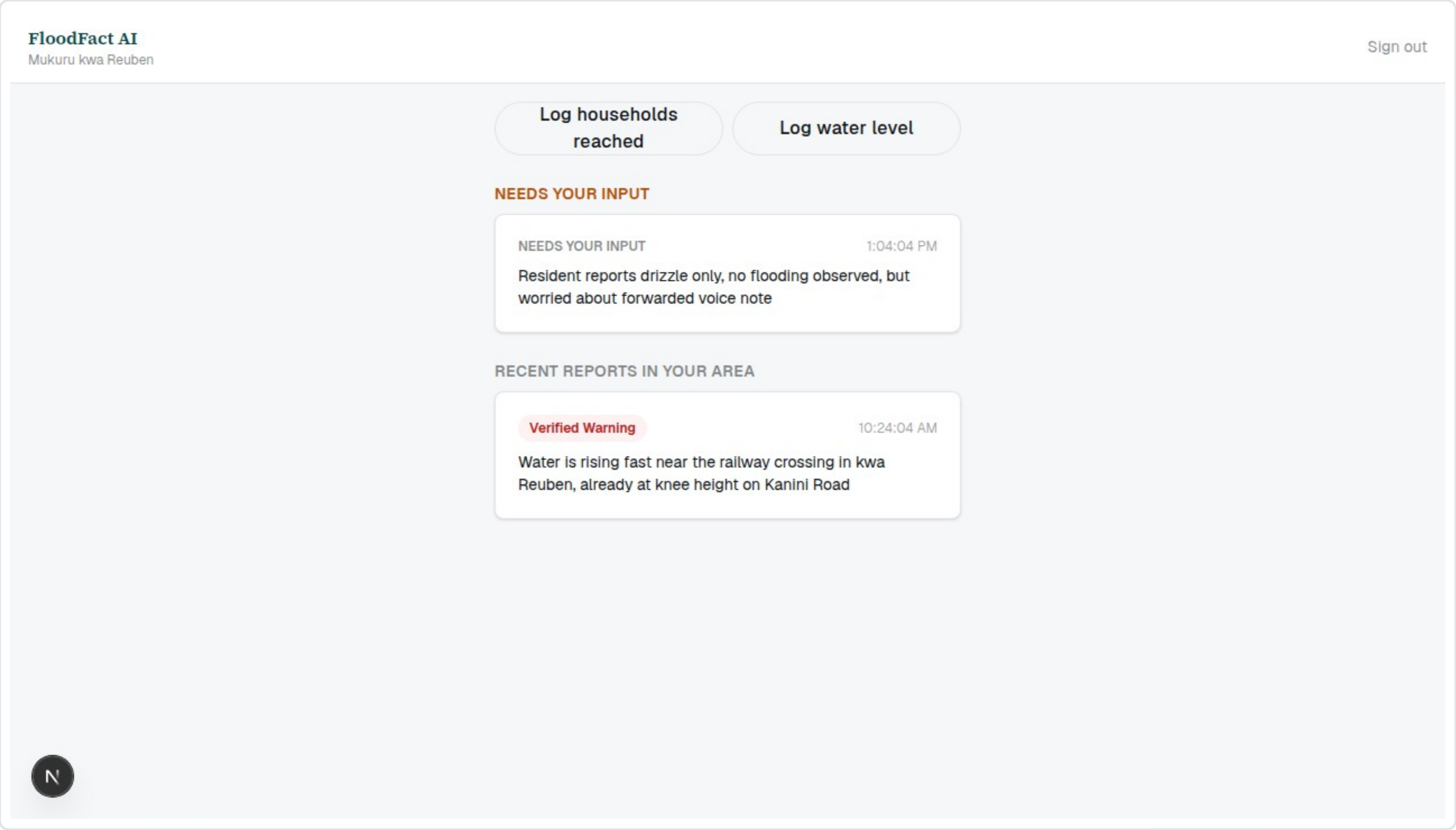
Live feed + status view, colored by verdict — Verified Warning, Elevated Risk, False Information.

Reporting takes seconds, even without WhatsApp



The same pipeline also runs on real inbound WhatsApp and SMS messages — this form is the fallback channel.

Ground truth from the people already trusted locally



Installable, offline-tolerant PWA — built for patchy 2G/3G coverage, not a desktop-first dashboard.

BUILT WITH REAL PARTNERS

Every integration in this build is real — not a mockup

Provider	Used for	Status
Esri ArcGIS	Flood-risk zone geography	Live-ready, sandbox fallback
Open-Meteo	Rainfall / ERA5 reanalysis	Live — free, keyless
Meta WhatsApp Cloud API	Resident intake + alerts	Live-ready, sandbox fallback
Africa's Talking	SMS intake + alerts	Live-ready, sandbox fallback
Anthropic Claude	Message understanding + rationale (never the decision)	Live-ready, sandbox fallback
Voyage AI + pgvector	Rumour-pattern semantic matching	Live-ready, lexical fallback
Supabase (Postgres/PostGIS/pgvector)	The entire backend	Live

Every provider auto-detects live vs. sandbox mode from its own credentials — no code changes needed to go live.

EXPECTED REAL-WORLD BENEFIT

Phase 1 pilot targets, tied to acceptance criteria

80%

of households receive alerts within 30 minutes

Pilot target

50%

reduction in rumour spread in pilot zones

Pilot target

100%

of youth ambassadors trained in the AI tool

Pilot target

Every one of these is computed live from recorded reports, alerts and ambassador logs (the pilot_metrics database view) — never a hand-typed demo number.

WHY FLOODFACT AI WINS

Unique selling points, at a glance

	Generic chatbot fact-checker	Alert-broadcast system	FloodFact AI
Evidence-based decision	✗ LLM opinion	✗ single source	✓ 6 sources, deterministic
Handles rumours	partial	✗	✓
Human trust layer	✗	✗	✓ youth ambassadors
Explainable verdict	sometimes	✗	✓ evidence-grounded
Real channels people use	✗	✓	✓ WhatsApp + SMS + notice points
Safe under uncertainty	✗ can hallucinate	n/a	✓ escalates to a human

ROADMAP

Built to scale beyond Mukuru without a rewrite

1

Phase 1 — Now

Flood + WhatsApp rumour verification. Mukuru: kwa Reuben, kwa Njenga, Viwandani.

2

Phase 2

Add landslide and fire hazards — new provider adapters and rows, same pipeline.

3

Phase 3

Expand to SMS, radio, TV, USSD, and national scale across Nairobi's settlements.

SDG ALIGNMENT

Grounded in the goals this hackathon serves

SDG 11 — Sustainable Cities

Safer, more resilient informal settlements in Nairobi.

SDG 13 — Climate Action

Community-led flood early warning and climate adaptation.

SDG 17 — Partnerships

KMD, Kenya Red Cross, UN-Habitat, youth groups, county government.

THANK YOU

Technology performs the evidence check. People carry the trust.

Full working system, source code, and setup guide: see the Demonstration file and the linked repository.